higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

INDUSTRIAL ELECTRONICS N5

4 AUGUST 2023

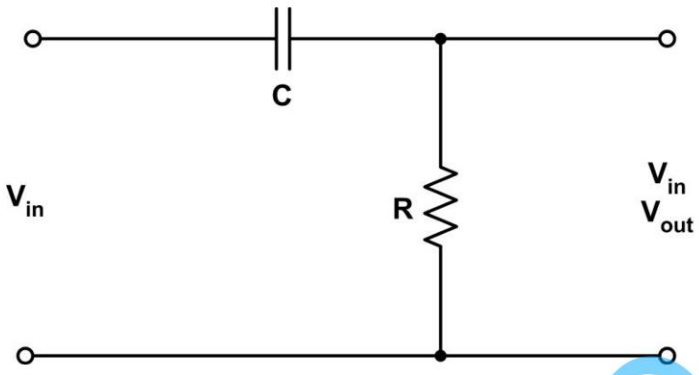
This marking guideline consists of 8 pages.

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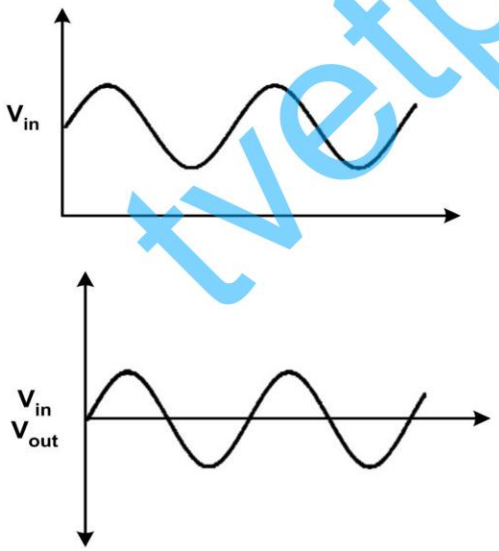
QUESTION 1

1.1

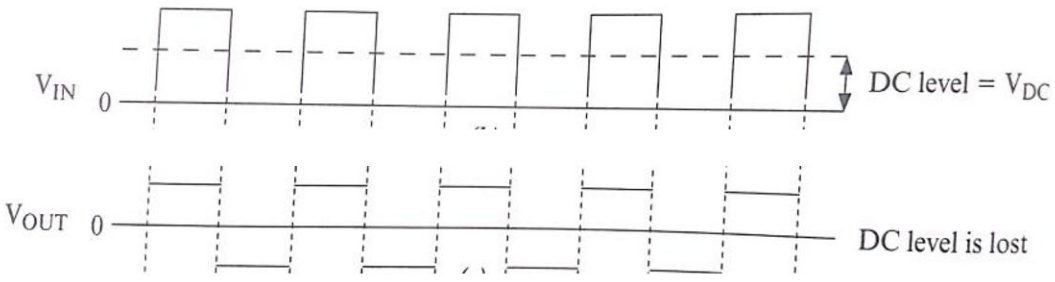


(1 mark for R and C labels; ½ mark for V_{in} and V_{out}) (3)

1.2



OR



(1 mark for V_{in} and V_{out} waveform) (2)

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1.3 A coupling circuit passes only the signal waveform from one part of the circuit to the next part of the circuit and blocks the DC component,✓ as it contains no information.✓ (2)

1.4 $f_r = \frac{1}{2 \times \pi \times (L \times C)^{1/2}}$

$= \frac{1}{2 \times \pi \times (0{,}2 \times 47 \times 10^{-6})^{1/2}}$ ✓

$= 51{,}911$ Hz ✓

$BW = \frac{R}{2 \times \pi \times L}$

$= \frac{100}{2 \times \pi \times 0{,}2}$ ✓

$= 79{,}577\ Hz$✓

$Lower\ frequency = f_r - \frac{BW}{2}$

$= 159{,}155 - \frac{79{,}577}{2}$ ✓

$= 12{,}122\ Hz$ ✓

$Upper\ frequency = f_r + \frac{BW}{2}$

$= 159{,}155 + \frac{79{,}577}{2}$ ✓

$= 39{,}789\ Hz$ ✓

(4 × 2) (8)
[15]

QUESTION 2

2.1
2.1.1 Control element/Series control element
2.1.2 Reference element
2.1.3 Comparator
2.1.4 Protection
2.1.5 Feedback element/Feedback network

(5 × 1) (5)

2.2 Series voltage regulator (1)

2.3 2.3.1 No-load voltage is that voltage that is supplied by the secondary winding of the transformer when the load to that power supply is not connected.

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2.2.2 Voltage regulation is defined as the change in the output voltage (full load) for varying load conditions.

2.2.3 A filter is a device or component that will remove the ripple (pulsating) component from the output of a rectifier circuit.

2.2.4 A voltage doubler is an electronic circuit that charges capacitors from the input voltage and switches these charges in such a way that, in the ideal case, exactly twice the voltage is produced at the output as at its input.

(4 × 2) (8)

2.4
- Mobile phones
- Laptops
- Charger circuits

(Any 2 × 1) (2)
[16]

QUESTION 3

3.1 3.1.1 $I_e = I_c = 10\ mA$

$$\therefore V_e = \frac{1}{10} \times V_{CC}$$

$$= \frac{1}{10} \times 12 \checkmark$$

$$= 1,2\ V \checkmark$$

$$R_C = \frac{V_{cc} - V_{ce} - V_e}{I_c}$$

$$= \frac{12 - 6 - 1,2}{10 \times 10^{-3}} \checkmark$$

$$= 480\ \Omega \checkmark$$ (4)

3.1.2 $$R_{b2} = \frac{1}{10} \times \beta R_e$$

$$= \frac{1}{10} \times 250 \times 120 \checkmark$$

$$= 3\,000\ \Omega \checkmark$$

$$V_b = V_e + V_{be}$$

$$= 1,2 + 0,6 \checkmark$$

$$= 1,8\ V \checkmark$$

$$R_{b1} = \frac{R_{b2}(V_{cc} - V_b)}{V_b}$$

$$= \frac{3\,000(12 - 1,8)}{1,8}$$

$$= 17\ k\Omega \checkmark$$ (6)

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3.2 $Zi = Rb \parallel Z1$

$= hie$

$= \underline{1{,}2\ k\Omega}$✓

$Z2 = \frac{1}{h_{oe}}$

$= \frac{1}{20 \times 10^{-6}}$

$= \underline{50\ k\Omega}$✓

$\frac{1}{Z_0} = \frac{1}{R_c} + \frac{1}{Z_2}$

$= \frac{1}{480} + \frac{1}{50 \times 10^3}$ ✓

$\therefore Z0 = \underline{475{,}436\ \Omega}$✓

OR

$Z_0 = \frac{R_c \times Z_2}{R_c + Z_2}$

$= \frac{480 \times 50\,000}{480 + 50\,000}$ ✓

$= \underline{475{,}436\ \Omega}$✓

(4)

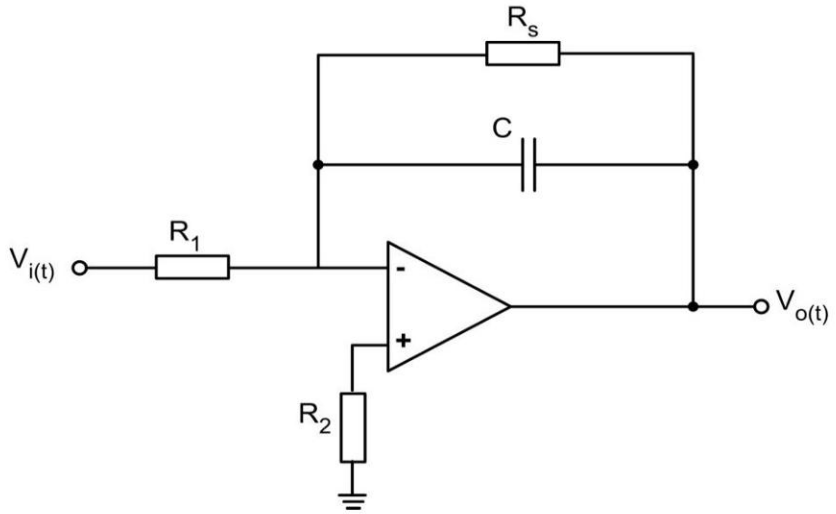
3.3 3.3.1 Collector current I_{co} doubles for each 10º in temperature.
3.3.2 The base-emitter voltage decreases with 2,5 mV/ºC.

(2 × 2) (4)
[18]

QUESTION 4

4.1 Drift is a slow variation of the work voltage or current✓✓ on the output while the input is kept at a constant level.✓ (3)

4.2



(1 mark for the OPAMP; ½ mark for V_{in}, V_{out}, R_1, R_2, C, R_s each correct component) (4)

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4.3 A voltage adder adds the voltages, of which each is multiplied with a constant gain factor, i.e. inputs can be two or more. (1)

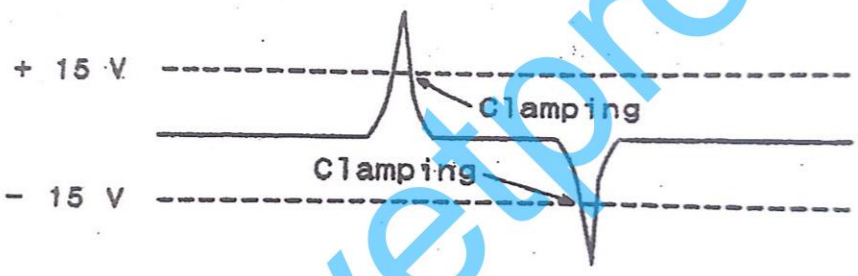
4.4
- Very high input impedance that causes almost no current load for circuits
- Very high voltage gain that is suitable to amplify very small signals
- Very low output impedance that is not affected easily by other loads (3)

[11]

QUESTION 5

5.1
- Resistor-diode network
- Double diode concept
- Triple diode concept
- Zener diode protection network (Any 2 × 1) (2)

5.2



(½ mark for +15 V and -15 V labels; 1 mark for positive clamping curve and 1 mark for negative clamping curve) (3)

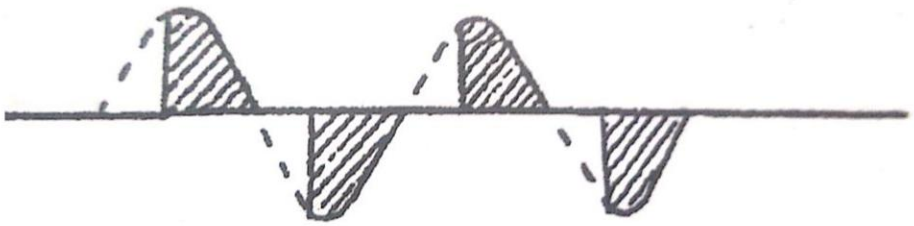
5.3
- Store unutilised units in conductive sponge/containers.
- Use earthed point soldering irons as soon as you solder the circuits.
- Use specially earthed wristbands.
- Connect all test equipment to earth.
- Switch off the power before working on the circuit.
- All unutilised inputs must be connected to V_{DD} or V_{SS}. (Any 2 × 1) (2)

[7]

QUESTION 6

6.1 The phase triggering circuit is responsible for the trigger pulse that is applied on the gate of an SCR, a triac or a quadraq.

6.2

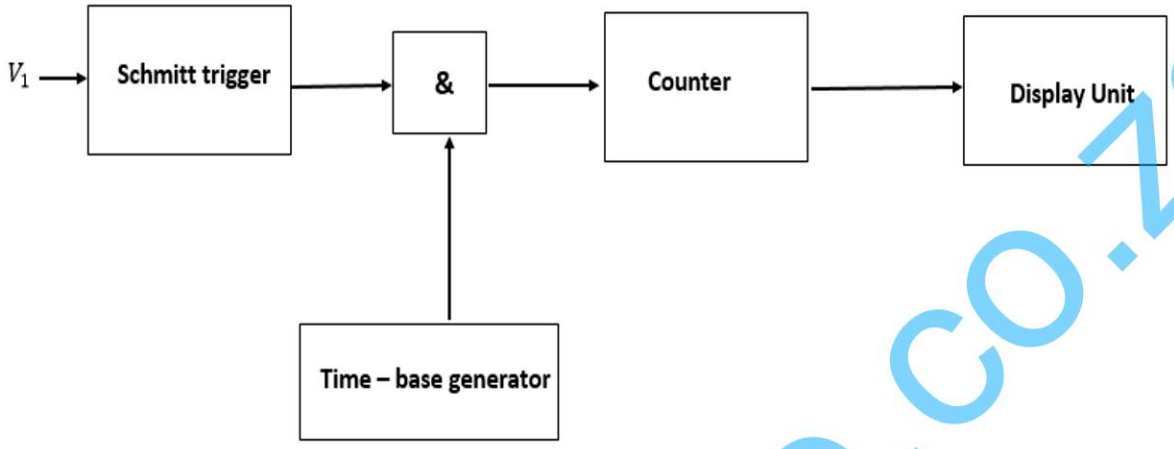


✓✓

(2 × 2) **[4]**

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QUESTION 7

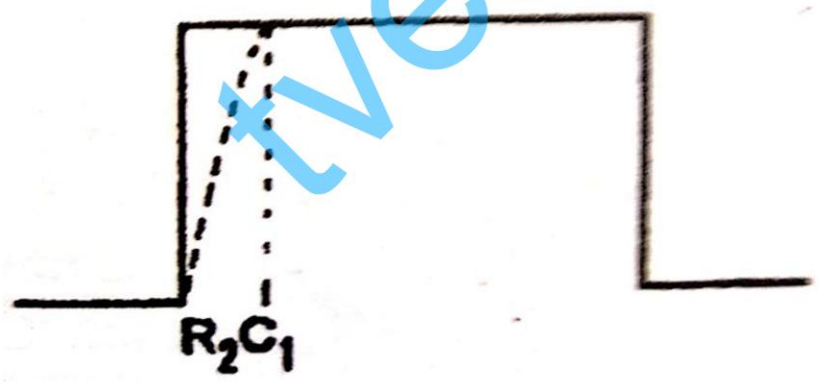


(1 mark for each correct block label) **[5]**

QUESTION 8

8.1 Rozner modification is used to improve the increase time. (2)

8.2



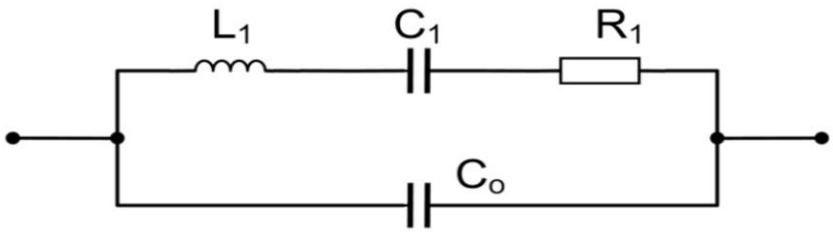
(½ mark for each correct label, R1 and C1; 1 mark for the square waveform) (2)

8.3 $$f = \frac{1{,}443}{(R_A + 2R_B)C}$$

$$= \frac{1{,}443}{(20 \times 10^3 + 2 \times 5 \times 10^3)\ 47 \times 10^{-6}}$$ ✓✓

$$= 1{,}023\ Hz$$✓ (3)

8.4



(½ mark for each correct component) (2)

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8.5
- The frequency the oscillator produces must be very stable.
- The amplitude of the output must be constant.
- Provision must be made for positive feedback.

(Any 2 × 1) (2)
[11]

QUESTION 9

9.1

9.1.1	G
9.1.2	C
9.1.3	F
9.1.4	A
9.1.5	B
9.1.6	D
9.1.7	E

(7 × 1) (7)

9.2 Passive transducers produce a variation in electrical parameters such as resistance, inductance and capacitance, which will be measured in current/ voltage.✓ OR
Passive transducers require electrical power to operate, e.g. resistive, inductive and capacitive transducers.

Active transducers will generate a voltage/current when subjected to some form of changing quantity and therefore do not require an external excitation supply.✓ OR
Active transducers generate an emf as soon as they are stimulated by some or other form of energy and therefore require no external supply. (2)

9.3 $R_{t1}: 12 \times 10^3 = Ae^{B/283}$ $-----(1)$ $(T = 273 + 10)$ ✓(1/2)

$R_{t1}: 2 \times 10^3 = Ae^{B/303}$ $-----(2)$ $(T = 273 + 30)$ ✓(1/2)

$(1) \div (2)$ $\frac{12 \times 10^3}{2 \times 10^3} = \frac{Ae^{B/283}}{Ae^{B/303}}$ ✓

$6 = e^{B/283 - B/303}$

$6 = e^{B(1/283 - 1/303)}$

$log\ 6 = B(2{,}33 \times 10^{-4}) \log e$ ✓

$B = \frac{\log 6}{(2{,}33 \times 10^{-4}) \log e}$

$B = -0{,}0914$ ✓ (4)
[13]

TOTAL: 100

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